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Values Are Not Enough: Situational Modifiers Shape Value-Based Decisions in LLMs and Humans

Key Highlights

- Research Problem: Values alone do not predict moral choices; situational modifiers can shift decisions in humans and LLMs.
- Proposed Solution: Create VISDA (80 scenarios, 8 modifier axes) and a decision-flip metric to measure context-induced choice changes across value profiles.
- Unique Contribution: Evaluate 3 LLMs, a utility baseline (95 Schwartz profiles), and humans; reveal LLM miscalibration favoring personal-cost modifiers.

Grammar Review

- By contrast, LLMs tend to be state-blind and respond mainly to trait-level prompts, and (Soni et al., 2025) reports that situational context can alter responses in ways dispositional profiles alone do not capture. - Subject-verb agreement error: with the plural subject "Soni et al." the verb should be plural ("report"), not singular ("reports").
- (Mahajan, 2025) encode responses to ethical dilemmas into an eleven-dimensional normative profile but acknowledge only limited framing variation across scenarios. - Subject-verb agreement error: a singular author name takes a singular verb ("encodes"), not plural ("encode").
- (Schacht and Lanquillon, 2025) provides a mechanistic analysis of ethical decision-making in LLMs, examining situational cues but without holding value profiles fixed. - Subject-verb agreement error: a compound plural subject requires a plural verb ("provide"), not singular ("provides").
- The 5 themes across which dataset construction is centered around are: 1. Resource allocation 2. Workplace 3. Recognition 4. Personal Choice 5. Leader Succession Scenarios. - Redundant prepositions ("across which ... centered around") create an ungrammatical construction; use only one prepositional structure, not both..
- Profiles To represent value orientations, we construct 95 curated Schwartz value profiles, each encoded as binary configurations over the ten Schwartz values. - Missing terminal punctuation after "Profiles" results in a fused sentence; a period is required to separate the heading-like element from the following sentence..
- 3.4 Utility-based Mathematical Baseline Decision-making is modeled through a simple utility-maximization baseline defined over the Schwartz value space. - Missing terminal punctuation between the section heading and the subsequent sentence creates a run-on; insert a period after the heading..
- All three models are deployed locally on cluster of 4 NVIDIA A100 GPU nodes and queried with greedy decoding (temperature = 0), which ensures deterministic outputs and reproducibility. - Missing indefinite article before a singular countable noun; it should read "on a cluster," not "on cluster."
- open-weight LLMs, a utility-based mathematical baseline, and a 50-participant human pilot, situational modifiers cause systematic decision flips that vary by axis, scenario, profile strength, and model scale. - The initial noun phrase is not syntactically integrated with the main clause, creating a comma splice and a dangling introductory phrase; the sentence lacks a coherent grammatical subject for the predicate..
- The human study was approved by IIT Hyderabad institutional review and conducted under informed consent with the right to withdraw. - The noun phrase "IIT Hyderabad institutional review" is incomplete; it lacks a required determiner and head noun (e.g., "the IIT Hyderabad Institutional Review Board/Committee"), rendering the construction ungrammatical..

Readability Review

- Among the most widely used frameworks is Schwartz's value theory (Schwartz et al., 2012), which organizes human values along higher-order dimensions such as openness-to-change versus conservation and self-enhancement versus self-transcendence. - Long sentence with a relative clause and two parallel comparisons packed into a single list, increasing processing load..

- Alternative frameworks such as Haidt's moral foundations theory (Haidt, 2012) capture broader moral intuitions, but Schwartz values map more directly onto action-level trade-offs, making them well suited for behavioral decision analysis. - Coordinates two independent clauses and adds a participial phrase, creating multiple clause layers in one sentence..
- For example, valuing security may favor cautious choices, while valuing benevolence may favor helping at personal cost; similarly, recent LLM studies show systematic behavioral shifts when value profiles are changed (Sauter and Schirmer, 2026; Guo et al., 2026). - Combines multiple contrasting clauses and a semicolon-linked clause, resulting in dense multi-part structure..
- Decades of social psychology, especially the person-situation debate (Harry et al., 2026; Soni et al., 2025; Peterson, 2025), suggest that behavior depends not only on stable traits but also on immediate contextual pressures. - Heavy fronted prepositional phrase with an embedded appositive and a not only...but also construction makes it syntactically complex..
- Few studies examine whether a fixed value profile yields different actions when only situational context changes (Harry et al., 2026; Sauter and Schirmer, 2026; Backmann et al., 2025), so it remains unclear whether LLMs act as stable value-driven agents or exhibit human-like context sensitivity (Harry et al., 2026; Backmann et al., 2025; Jiaqi et al., 2025). - Long multi-clause sentence with two embedded whether-clauses and extensive parenthetical details joined by a coordinating conjunction..
- As shown in Figure 1, each scenario pairs two actions with modifier sentences (e.g., emotional pressure, personal cost, social expectations, stakes), and situation-driven change is measured via a decision-flip metric. - Introductory clause plus coordination and a parenthetical list make the structure lengthy and layered..
- Social psychology has long debated whether stable dispositions or situational context better predict behavior (Mischel, 1968; Ross and Nisbett, 1991); recent work reports that a large share of variance is within-person (e.g., 74% in Harry et al., 2026). - Two independent clauses joined by a semicolon with embedded complement clauses and parentheticals increases complexity..
- More directly, Sauter et al., Backmann et al., and Nabizadeh et al. find that contextual framing, opponent behavior, survival risk, and broader ethical themes can all change moral judgments even when the underlying model is unchanged (Sauter and Schirmer, 2026; Backmann et al., 2025; Nabizadeh et al., 2026). - Long subject with multiple coordinated items and a subordinate even when clause yields a dense structure..
- Together, these results indicate that situational modifiers independently shape moral decisions beyond what value profiles alone predict; Peterson frames this as person-situation dynamics in LLM alignment (Peterson, 2025), echoing classic social-psychology findings that behavior depends jointly on stable dispositions and immediate context. - Contains a semicolon linking two clause clusters plus an appositive and a final that-clause, making it multi-layered..
- (Jiaqi et al., 2025) find that LLM personality traits are dynamic and input-driven rather than stable, suggesting that current evaluation designs may underestimate situational sensitivity. - Main clause followed by a participial clause of result adds complexity and length..
- Each theme additionally specifies a set of potential_{v}alue_{t}ensions, i.e., opposing value dimensions that may come into conflict within the scenario and are intended to be surfaced through the responses. - Long noun phrase with an explanatory appositive and two relative clauses increases cognitive load..
- Each scenario includes a unique scenario_{i}d, a corresponding theme_{i}d, a free-text description, an actor, two alternative actions (A0 and A1), and a latent_{v}alue_{t}ensions list inherited from the parent theme. - Lengthy coordinated list within a single sentence makes parsing the structure demanding..
- Each modifier further contains a modifier_{t}ext sentence inserted into the scenario and an expected_{v}alue_{p}ressure list specifying the Schwartz value dimensions it is intended to influence. - Multiple embedded modifiers (participial and relative clauses) and coordination create a dense construction..
- Second, action pairs were written to express different value priorities rather than different outcome quality, such as innovation vs. tradition, individual recognition vs. group recognition, or local security vs. broader welfare. - Uses contrastive phrasing and a long illustrative list, producing a complex sentence flow..
- From the remaining pool, we apply a coverage-balanced sampling heuristic stratified by profile strength ($||VSW||_{\{1\}}$) and higher-order quadrants, selecting profiles to maximize Hamming diversity while maintaining balanced marginal value frequencies. - Long sentence with multiple stacked modifiers (prepositional phrase, parenthetical detail, and participial clause) makes interpretation effortful..
- To verify that decision flips are not simply due to prompt-paraphrase instability, we evaluate four meaning-preserving paraphrases for a stratified sample of 95 baseline ($V_{\{SW\}}$, scenario) cells per model. - Introductory purpose clause plus a complex object with stacked modifiers and parentheticals increases complexity..
- The utility-based mathematical baseline instead ranks authority signal and in-out-group highest, indicating a divergence from LLM behavior: learned models are comparatively more sensitive to direct authority and self-threat signals than to identity-based modifiers. - Main clause followed by a colon introducing an explanatory clause with comparative phrasing creates a heavy, multi-part structure..
- To test whether each modifier produces directional rather than random flips, we apply McNemar's exact test on the paired baseline-vs-modifier outcomes for each (model, axis) cell, with Benjamini-Hochberg FDR correction across the eight axes. - Purpose clause leads into a long methodological description with compound prepositional phrases, increasing density..
- Our three LLMs are trained by different organizations on different data and alignment recipes - Gemma 4 31B (Google DeepMind), Qwen 2.5 32B (Alibaba), LLaMA 3.1 8B (Meta) - so agreement on axis ordering serves as independent replication. - Long sentence with interrupting dashes and an extensive appositive list adds structural complexity..

- Spearman correlation on the full eight-axis ordering gives $\rho = 0.66$ between the two $\geq 30B$ models ($p = 0.076$), with LLaMA 8B sitting lower (Qwen $\rho = 0.49$, Gemma $\rho = 0.18$); the utility-based baseline is weakly or negatively correlated with every LLM ($\rho \in [-0.13, 0.35]$) despite itself flipping on 19.93% of trials. - Two semicolon-linked clauses with multiple parenthetical inserts and comparative structures make it lengthy and hard to follow..
- Cross-pipeline convergence on the same two axes rules out the most natural competing explanation - that the effect is an artifact of one lab's RLHF data - while the divergence from the additive utility rule indicates that LLMs are not implementing a noisy version of value-arithmetic. - Complex sentence with two coordinated main clauses, interrupting parenthetical set off by dashes, and a long final that-clause..
- When a value profile contains 8 or 9 of 10 active values, the utility-based mathematical baseline is locked by construction: the active-value dot product dominates the modifier perturbation, so the baseline flips on zero such trials. - Long conditional setup followed by a colon and a coordinated explanation increases structural complexity..
- The term is baseline-relative: such flips are impossible under the specified value-arithmetic rule, but remain possible for a model that treats the modifier as a contextual premise rather than as a fixed value increment. - Uses a colon, contrastive conjunction, and an extended comparative phrase, creating multiple layers..
- In a recognition scenario, Gemma initially selects team recognition for a profile high in universalism and benevolence; when the modifier states that management prefers measurable results, it switches to individual recognition and cites that preference. - Two clauses joined by a semicolon with a dependent when-clause and multiple prepositional phrases creates a dense structure..
- For alignment, the implication is mixed: non-additivity reduces the excessive flipping predicted by the utility baseline, but it can also create axis-specific vulnerabilities when models overweight cues such as authority or self-preservation relative to human reference behavior. - Introductory phrase and colon followed by a long contrastive clause with embedded when-phrase increases complexity..
- We group the eight modifier axes into four pressure types to analyze whether different categories of situational pressure produce systematically different decision-shift patterns: stakes (resource_{s}carcity), affective (authority_{s}ignal, social_{v}isibility, in_{o}ut_{g}roup), personal-cost (self_{p}reservation, time_{p}ressure), and informational (diffused_{r}esponsibility, competence_{u}ncertainty). - Long sentence culminating in a colon and a multi-part taxonomy list makes it heavy to parse..
- The human pilot (N = 50, 500 decisions) shows a pooled between-subjects decision shift of 12.36% (Table 1), placing humans well above LLaMA 3.1 8B (2.03%) and above Qwen 2.5 32B (6.58%), in the same range as Gemma 4 31B (8.92%), and below the additive utility baseline (19.93%). - Main clause followed by a long participial phrase with multiple comparative inserts and parentheticals increases complexity..
- Open-weight LLMs, a utility-based mathematical baseline, and a 50-participant human pilot, situational modifiers cause systematic decision flips that vary by axis, scenario, profile strength, and model scale. - Long sentence with multiple fronted noun phrases and a heavy that-clause listing several dimensions reduces readability..
- The two axes that most reliably move LLMs—authority signal and self-preservation—are recognizable analogues of two of the most influential situational factors in classic social-psychology experiments (obedience and self-interest). - Extended subject with embedded dashes and a parenthetical example creates a lengthy nominal structure..
- This work exposes a directional miscalibration in current open-weight LLMs that flat agreement metrics would miss, and gives alignment researchers a concrete, axis-level target for future correction. - Coordinates two main predicates and embeds a relative clause, making the sentence lengthy and multi-clausal..
- Data was collected from a convenience sample of college students, staff, and acquaintances, sharing a single educational and cultural context; broader replication is necessary before generality claims. - Semicolon joins two clauses and the first clause contains a long prepositional phrase and participial modifier, increasing complexity..
- We use the classic Schwartz 10-value framework rather than the 19-value refined theory (Schwartz et al., 2012); fine-grained distinctions (e.g., humility, face, societal vs. interpersonal conformity) collapse into broader categories and may obscure value-specific modifier effects. - Two clauses linked by a semicolon with an extensive parenthetical list yields a complex, layered structure..
- Real-world values are continuous; our binary encoding follows precedent (Sorensen et al., 2024) but coarsens the signal, and pruning antagonism-violating combinations further constrains the profile space. - Semicolon-connected clauses with coordination and nominalizations create a dense and multi-part structure..

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